AAE 518: Zero-gravity Fluid Dynamics

Credit Hours	3
Offered	Fall
Pre-requisites	None
Co-requisites	None
Instructional Method	3 hours of lecture per week
Required	No

1. Course Description

Introduce students to a variety of fluids topics relevant to the weightless periods of spaceflight. Basic fluid physics, history of design and applications, current research topics, and terrestrial spin-offs are all topics during the semester. Because the field of zero-gravity fluids is such a small community, it is assumed that few if any of the students in AAE518 will have a career in zero-gravity fluids. Thus, the course seeks to teach fluid physics and the application of this knowledge to diversify fluids specialists and to enhance the systems knowledge in future satellite propulsion and GN&C engineers. Numerical modeling is not traditional CFD but the complementary and mildly peculiar *Surface Evolver* code for capillary fluid statics and stability.

2. Instructor Information

Prof. Steven Collicott
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3. Topics Covered

1. Examples of zero-gravity and zero-gravity basics.
2. What is zero-gravity, weightlessness, etc?
    a. High school physics review
    b. Orbits review
    c. Simple orbit perturbation and gravity gradient effects
3. Fluid properties, such as surface tension and contact angle
    a. Definitions and discussion
    b. Terrestrial examples
4. Energies in capillary fluids problems
    a. Non-dimensionalization
    b. Scaling
5. Capillary Stability:
    a. classical examples
    b. recent applications
6. Three-dimensional solutions: the *Surface Evolver* code.
    a. Demos
        i. Running the demos
        ii. Vector calculus – applications of the Divergence Theorem and Green's Theorem to the real world.
    b. Defining your own geometries in *Surface Evolver.*
        i. Starting wisely
        ii. Defining clearly
        iii. Debugging effectively
        iv. Getting information out of the code and into use.
7. Critical contact angle phenomena
    a. Examples
    b. Concus-Finn analysis
    c. *Surface Evolver* use
8. Capillary-driven corner flows – imbibition
    a. Analysis
    b. Experiment
    c. Application
9. Droplets and bubbles in tubes or micro-channels
    a. Applications
    b. Solving multiple topologies

10. Projects
    a. Goals
    b. Topics
    c. Progress Reports
    d. Presentations

4. Intended Learning Outcomes

On completing this course, the student shall be able to:

- Understand and analyze the effects of scale on mixed gravitational and capillary fluids problems.
- View a new capillary fluid statics geometry and predict the relative pressures in the different phases.
- Apply results of critical wetting analysis to design practical fluid systems to either transport liquid or capture liquid in one place in weightlessness.
- Apply results of imbibition analyses to predict the speed of capillary advance in grooves.
- Create numerical models of spacecraft two-phase fluids systems for analyzing propellant control in satellite tanks.
- Describe results of their modeling and analysis to other engineers.

5. Assessment Method

Regular homeworks, a mid-term exam, and a small-team project.

6. Relation to ABET Outcomes

	Program Learning Outcomes	Included?
1.	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	Yes

2.	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	Yes
3.	An ability to communicate effectively with a range of audiences.	Yes
4.	An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	Yes
5.	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	Yes
6.	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	No
7.	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	Yes